INVOICE PAYMENT PREDICTION

Exploratory Data Analysis Report

Records analysed: 1,048,575
EDA columns: 37
Scope: Payment status, payment timing, invoice amounts, customer behaviour, time trends, aging proxy, and feature relationships

This report summarizes the completed exploratory analysis. Monetary amounts in the dataset span multiple currencies and must not be interpreted as a single-currency total.

1. Executive Summary

The analysis covered 1,048,575 training invoice records. The dataset contained 31 original fields and six derived EDA fields. All six derived fields passed validation against their source dates. The EDA dataset had zero missing values.

Measure	Observed result
Total invoices	1,048,575
Paid	713,456 (68.04%)
Pending	178,137 (16.99%)
Overdue	104,373 (9.95%)
Cancelled	52,609 (5.02%)
Outstanding (Pending + Overdue)	282,510
Paid invoices with positive delay	0

Main findings

• Paid records represent 68.04% of invoices; Pending and Overdue together represent 26.94%.
• Of paid invoices, 96.03% are recorded as early and 3.97% on time; none are late.
• Payment terms have a notable negative association with payment delay (Pearson approximately -0.5865).
• Invoice amount has almost no observed association with payment delay (Pearson approximately 0.0006).
• Average payment delay remains close to -15 days across countries and months.

Critical limitation: there are no paid-late examples in the dataset. A model trained on these records cannot learn a reliable paid-late pattern.

2. Dataset and Preparation

The six derived columns are payment_terms_days, invoice_month, invoice_year_month, invoice_weekday, invoice_day, and is_month_end. The full local EDA file was saved as Parquet. An anonymized public CSV sample contains 10,486 rows and 27 columns after direct identifiers were removed.

3. Payment Behaviour and Relationships

Payment timing

Among 713,456 paid invoices, 685,128 (96.03%) were paid early and 28,328 (3.97%) on the due date. No paid invoice had a positive delay. Non-paid records have payment_date equal to due_date in the supplied data; this should be treated as a placeholder, not evidence of an actual payment.

Invoice amount and payment terms

Relationship	Statistic	Interpretation
Invoice amount vs payment delay	Pearson 0.0006; Spearman 0.0005	No meaningful observed association
Payment terms vs payment delay	Pearson -0.5865; Spearman -0.5199	Longer terms associated with more negative delay
Subtotal vs total amount	Pearson 0.988	Strong formula-related relationship
Tax amount vs total amount	Pearson 0.860	Strong relationship
Subtotal vs tax amount	Pearson 0.803	Strong relationship

Mean payment delay by terms was approximately -7.50 days for 15–21 day terms, -15.01 for 22–30, -22.64 for 31–45, and -29.96 for 46–60. These are associations, not proof of causation.

Country and repeat-customer patterns

Average delay across countries was consistently close to -15 days. Among 45,048 customers with at least 10 paid invoices, the average early-payment share was 96.03%; late-payment share was zero. Historical customer features must be calculated using only invoices earlier than the invoice being scored.

4. Time Trends and Aging Proxy

Time trends

The dataset covers 60 invoice months. Average monthly volume was approximately 17,476 invoices. The highest month was August 2021 (18,202 invoices); the lowest was February 2022 (15,895). Average monthly paid-invoice delay remained close to -15 days.

Month-end analysis should compare average daily counts, not raw totals, because the groups contain unequal numbers of days. The previously calculated averages were approximately 576.45 invoices per month-end day and 573.81 per regular day, a small difference.

Aging analysis — qualification

The aging proxy used 2024-02-29, the latest due date in the dataset, as its reference date. Outstanding records were defined as Pending plus Overdue (282,510 records). Days overdue were calculated as reference date minus due date.

Aging bucket	Count	Share
0–30 days	910	0.32%
31–60 days	3,892	1.38%
61–90 days	4,669	1.65%
90+ days	273,039	96.65%
Total	282,510	100.00%

This is not a verified current receivables aging snapshot. About 63% of each aging bucket is labelled Pending, including records more than 90 days past due under this calculation. Treat it only as a dataset-based proxy.

Currency limitation

The dataset contains multiple currencies. Combined amount totals and distributions are raw numeric values across currencies, not a single-currency financial balance. Group by currency or convert using documented exchange rates before financial interpretation.

5. Insights, ML Hypotheses, and Recommendations

#	Chart-backed insight	Business / ML interpretation
1	Status: Paid 68.04%; Pending 16.99%; Overdue 9.95%; Cancelled 5.02%.	Define status task carefully and exclude leakage fields.
2	Paid timing: 96.03% early; 3.97% on time; 0% late.	Acquire genuine late-paid examples before training a late-payment classifier.
3	Amount-delay correlation is approximately zero.	Test amount only with currency-aware features.
4	Payment terms-delay Pearson correlation is -0.5865.	Test payment terms with chronological validation.
5	Country average delays are close to -15 days.	Test whether country adds predictive value.
6	Repeat customers average 96.03% early payments.	Build time-aware customer-history features.
7	Aging proxy: 96.65% falls in 90+ days.	Do not treat this as verified operational aging.
8	Monthly average delay stays near -15 days.	Test calendar features using an out-of-time holdout.
9	Subtotal-total correlation is 0.988; month-end daily difference is small.	Avoid redundant formula fields; test month-end effects.

Recommendations before modelling

1. Define the target and prediction timestamp precisely.
2. Exclude payment_date and is_overdue from invoice-time prediction features; audit all fields for leakage.
3. Obtain genuine paid-late examples.
4. Use chronological train/validation/test splits.
5. Build customer history only from earlier invoices.
6. Group monetary analysis by currency or convert with documented rates.
7. Replace the aging proxy with a verified reporting date and reliable payment-state history.

Conclusion

The dataset shows predominantly early payment among paid invoices and a notable association between payment terms and payment delay. Invoice amount has almost no observed relationship with delay. Country and monthly delay patterns are stable. However, the lack of paid-late examples, placeholder-like payment dates for non-paid records, multiple currencies, and aging/status inconsistencies are material constraints. These must be addressed before claiming that a late-payment prediction model is reliable.

Report scope: This PDF summarizes previously computed EDA statistics. It is not an audit of source-system financial records or a model evaluation report.